Jazz Graph:
A Graph Neural Network for Jazz Music Recommendation

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Problem Statement

Create musical recommendation by encoding collaboration as similarity signal?

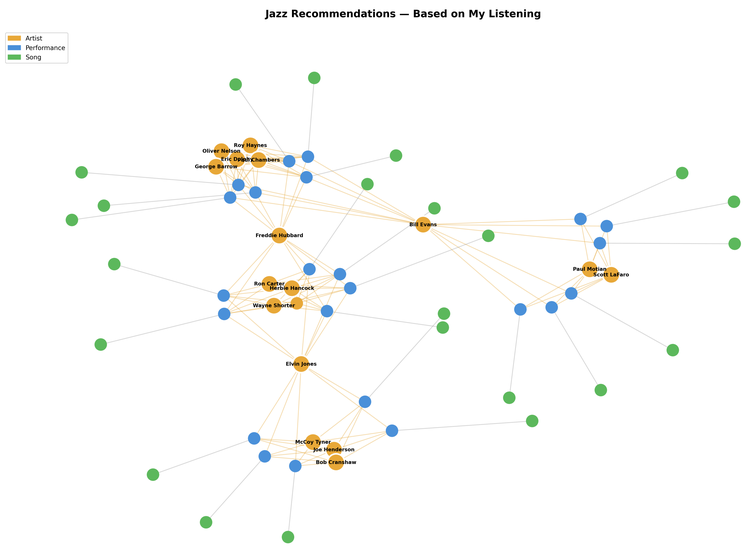

Figure: A sub-graph illustrating connections among jazz performances.

Background and Related work

Explain what you did.

System Architecture

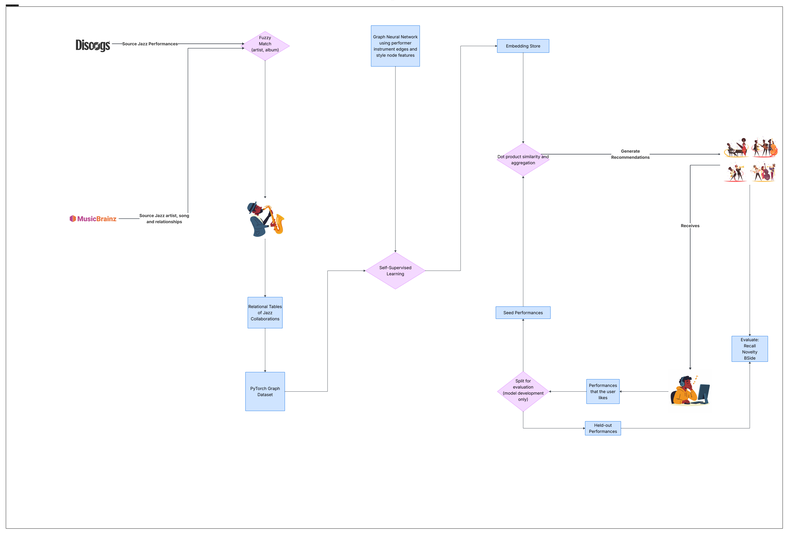

Figure: A data extraction pipeline(left), self-supervised learning with GNN (middle) and recommendation system (right.)

Results

- Novel Recall: Relevance of model recommendations.
- Novelty: Exploratory model behavior.
- B-Side: Expected recommendations for narrow seeds.

Task	Features	Pooling	Novel Recall	Familiar Recall	Novelty	B-side Mean
Combined Loss	Instrument, Style	sum	0.276	0.328	0.806	0.822
Combined Loss	Instrument, Style	max	0.297	1	0.840	**0.839**
Combined Loss	Instrument, Style	softmax	0.276	0.318	0.807	0.822
Edge Ablation	Instrument, Style	sum	0.314	0.365	0.910	0
Edge Ablation	Instrument, Style	max	0.239	1	**0.917**	0
Edge Ablation	Instrument, Style	softmax	0.323	0.393	0.910	0
Match Album	None	sum	0.278	0.378	0.850	0.434
Match Album	None	max	0.173	1	0.835	0.345
Match Album	None	softmax	0.314	0.395	0.840	0.411
Match Album	Instrument, Style	sum	0.485	0.544	0.805	0.548
Match Album	Instrument, Style	max	0.295	1	0.829	0.523
Match Album	Instrument, Style	softmax	**0.518**	**0.608**	0.804	0.548
Random Walk Baseline	—	—	0.194	0.202	0.523	0.132
Artist Weighted Baseline	—	—	0.420	0.405	0.011	0

Conclusion

Key takeaway here.

Challenges and Limitations

Key takeaway here.

Future Work

Key takeaway here.

Discovery

Figure: Lester Bowie, "The Great Pretender" (1982) Great music I discovered with JazzGraph.

Repository

Figure: https://www.github.com/philosofool/jazz-graph